GraniteMeasure AI

Comprehensive User Walkthrough

For the Granite Department

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Executive Summary

GraniteMeasure AI is an intelligent granite block measurement and analysis platform that transforms traditional granite quarry operations through digital transformation.

The platform enables:

- Capture granite block dimensions using smartphone cameras
- Measure dimensions with centimeter-level accuracy using AI-powered photogrammetry
- Verify measurements through multi-view 3D reconstruction
- Optimize cutting patterns and estimate usable yield
- Assess seigniorage and tax obligations automatically

Why Block Measurement Matters

For granite quarry departments, accurate block measurement is the **core pain point**:

- Traditional tape-measure methods are error-prone and subjective
- Disputes over dimensions lead to revenue leakage
- Manual recording creates audit trail vulnerabilities
- No digital record for blockchain-style governance
- Cutting decisions made without accurate geometry

GraniteMeasure AI solves these problems through:

- Evidence-based digital twin creation
- Multi-view photogrammetric reconstruction
- Automated, auditable measurement records
- AI-powered yield optimization

1. Introduction

1.1 What is GraniteMeasure AI?

GraniteMeasure AI is a mobile-first platform that enables quarry operators, government officials, and stone processors to capture, measure, and analyze granite blocks with unprecedented accuracy and reliability.

1.2 Why Block Measurement Matters

Accurate measurement is the foundation of granite governance. Without it, departments face revenue leakage, disputes, and lack of accountability. GraniteMeasure AI provides evidence-based measurements that serve as the single source of truth.

2. System Architecture Overview

2.1 Platform Components

The platform consists of three main layers:

User Interface (Frontend): Next.js Web Application, Mobile-optimized PWA, Android Native App, Telugu + English Localization

API Gateway (Backend): FastAPI REST API (Port 8000), PostgreSQL Database, Redis for Caching, CORS-enabled

AI Analysis Engine: Multi-View Photogrammetry, Granite Detection (YOLOv8), Segment Anything, Defect Detection, Calibration Methods, Geometry Reconstruction

2.2 Data Flow

1. BLOCK CREATION → User registers block with quarry, category, GPS
1. CAPTURE SESSION INITIATION → Session created with calibration method
1. EVIDENCE CAPTURE → User captures 10+ photos with GPS metadata
1. IMAGE VALIDATION → AI checks quality, granite detection, duplicate detection
1. ANALYSIS JOB SUBMISSION → Submission triggers background analysis
1. 3D RECONSTRUCTION → Multi-view processing calculates dimensions
1. REPORTING & VERIFICATION → Results displayed for manual verification

3. Core Workflow: Block Measurement

(The Department's Primary Pain Point)

Step 1: Block Registration

Access: **/blocks** → **'New Analysis'** button

- User selects Quarry from registered quarry network
- User selects Granite Category (e.g., Red Galaxy, White Galaxy)
- User enters Block Code (auto-generated or manual)
- User captures GPS coordinates (optional but recommended)
- System creates permanent block record with unique ID
- Block status set to: REGISTERED

API Endpoint:

```
POST /api/v1/granite-blocks { "block_code": "GR-2024-001", "quarry_id":
"uuid-here", "granite_category_id": "uuid-here", "capture_date":
"2024-10-01T10:00:00Z", "gps": {"latitude": 17.3850, "longitude": 78.4867,
"accuracy": 5.0}, "notes": "North face quarry, phase 3" }
```

Step 2: Capture Session Setup

The system creates a capture session with one of three calibration methods:

- **MANUAL_REFERENCE**: User specifies known reference size and pixel width (quick, no markers)
- **ARUCO**: Detects Aruco markers in scene (controlled environments)
- **MEASUREMENT_BOARD**: Uses calibrated board with known dimensions (high precision)

Step 3: Guided Evidence Capture (The Heart of Accuracy)

Most measurement systems just take 'some photos.' GraniteMeasure AI uses **AI-guided capture** to ensure complete 360° evidence:

Required Core Views (5 minimum):

- 1. FRONT - Main visible face
- 2. BACK - Rear surface
- 3. LEFT - Complete left side
- 4. RIGHT - Complete right side
- 5. TOP - Elevated/top view

Bridge Views (AI-requested if overlap weak):

- 6. FRONT_RIGHT - Oblique 30-45° right
- 7. BACK_RIGHT - Oblique rear-right
- 8. BACK_LEFT - Oblique rear-left
- 9. FRONT_LEFT - Oblique front-left

- 10. BOTTOM - Only if directly visible

Validation Checks Performed:

- GRANITE DETECTION - YOLOv8 detector must detect granite
- QUALITY ASSESSMENT - Blur, exposure, noise checks
- CALIBRATION VERIFICATION - Reference object visibility
- DUPLICATE DETECTION - Perceptual hashing against existing
- OVERLAP VERIFICATION - Feature matching with existing images
- VIEW MISMATCH DETECTION - Verify correct side uploaded

Step 4: Complete Capture & Submit for Analysis

Minimum requirements: **5 accepted images** with full coverage.

Step 5: AI-Powered 3D Reconstruction & Measurement

The analysis pipeline performs:

1. VALIDATING IMAGES (5+ accepted required)
2. DETECTING GRANITE (YOLOv8 on each image)
3. SEGMENTING GRANITE (Create masks)
4. CHECKING CALIBRATION (Calculate cm/pixel)
5. MATCHING FEATURES (ORB/SIFT extraction)
6. ESTIMATING CAMERA POSES (Structure from Motion)
7. RECONSTRUCTING GEOMETRY (Create point cloud)
8. ESTIMATING DIMENSIONS (Extract L×B×H)
9. CALCULATING VOLUME (0.812 m^3 typical)
10. ANALYZING DEFECTS (Cracks, holes, patches)
11. OPTIMIZING CUTTING (3 orientations evaluated)
12. ESTIMATING WASTAGE (Waste percentage calculated)

Step 6: Viewing Analysis Results

- RECONSTRUCTION - Interactive 3D point cloud viewer
- MEASUREMENTS - Dimensional analysis with confidence scores
- DEFECTS - Detected cracks, holes, patches
- CUTTING - Cutting optimization recommendations
- EVIDENCE - Gallery of all captured images

- REPORT - Generate PDF report

Step 7: Manual Verification

High-value blocks or low-confidence measurements trigger manual verification.

Step 8: Seigniorage Assessment

Formula: **Seigniorage = Volume (m^3) × Rate (per m^3) × Density Factor**

4. Technical Deep Dive: Block Measurement Accuracy

4.1 Measurement Uncertainty Sources

Calibration Accuracy: Manual: ±1-2 pixels, Aruco: ±0.5-1%, Board: ±0.3-0.8% *(Combined: ~1-2% of measurement)*
Photogrammetry Error: Feature matching: ±0.5-1 pixels, Camera pose: ~1-3% *(Combined: ~2-3%)*
User Capture Error: Incomplete coverage: Up to 10%, Poor angle: 5-15% *(Best practice (8-10 views): <2%)*
Total Expected Uncertainty: Well-executed: ±1-2 cm (1-2%), Average: ±2-3 cm (2-3%) *(Poor execution: ±5+ cm (5%+))*

4.2 Confidence Scores Explained

- Number of accepted images (more = higher confidence)
- View distribution (360° = higher than partial)
- Calibration quality (clear reference = higher)
- Feature match count (more matches = higher)
- Reconstruction consistency (low reprojection error = higher)

Interpretation:

- 0.95-0.99: Excellent - multiple high-quality views
- 0.85-0.95: Good - adequate coverage
- 0.70-0.85: Acceptable - flag for review
- < 0.70: Poor - review recommended

5. Performance Benchmark

5.1 Processing Speed

On Device (Mobile): Image upload: 2-5 seconds, Validation: <1 second, Full analysis: 30-90 seconds

On Backend Server: 5-6 images: 30-45 seconds, 8-10 images: 45-70 seconds, 10+ images: 60-90 seconds

5.2 Accuracy Benchmarks

- Length: ±1.0-1.5 cm (0.7-1.5%)
- Breadth: ±0.8-1.2 cm (1.0-1.8%)
- Height: ±0.5-1.0 cm (0.8-1.6%)
- Volume: ±2-3% (typical)

6. Best Practices for Department Users

6.1 Best Practices for Accurate Measurement

PRE-CAPTURE:

- Inspect block before capture - identify best viewing angles
- Clear obstacles from around block (people, tools, vehicles)
- Check lighting - avoid direct sun glare or deep shadows
- Enable GPS on device (for field audit trail)
- Prepare reference object if using manual calibration

DURING CAPTURE:

- Capture all 10 standard views (don't skip 'easy' sides)
- Keep camera steady - use both hands or tripod
- Fill frame with block - minimal empty space
- Maintain good overlap between adjacent views (30-50%)
- Capture at consistent distance (1-2 meters typical)
- Use 4K/1080p resolution (not compressed thumbnails)

POST-CAPTURE:

- Review validation results - retake rejected images
- Ensure minimum 5 accepted images (aim for 8-10)
- Check 360° coverage - no large gaps
- Submit for analysis only when all validation green
- Record field notes for unusual block features

6.2 Common Pitfalls to Avoid

Insufficient Views: Only 2-3 sides captured. Impact: AI cannot reconstruct 3D geometry. *Solution: Solution: Capture all 10 standard views*

Poor Lighting: Strong shadows or blown-out highlights. Impact: AI cannot detect granite edges. *Solution: Solution: Capture during morning/evening*

Wrong Calibration: Using Aruco without actual marker. Impact: Calibration fails. *Solution: Solution: Match calibration type to conditions*

Occlusion: People, vehicles in frame. Impact: AI detects non-granite objects. *Solution: Solution: Clear scene before capture*

Moving Block: Block moves between images. Impact: Feature matching fails. *Solution: Solution: Capture in stable position*

7. Audit Trail & Compliance

7.1 Complete Audit Log

Every action is logged in the audit_logs table:

- block_created (with block_code)
- image_captured (with side)
- image_validated (accepted/rejected with reasons)
- capture_session_created
- capture_session_complete
- analysis_started
- analysis_completed / analysis_failed
- manual_verification_submitted
- seigniorage_assessed

7.2 Compliance Features

- Immutable block registration (can't alter block_code)
- GPS timestamp for each capture
- Operator ID tracking for accountability
- Image versioning (original files stored)
- Analysis job history (all state transitions)
- Manual verification records
- Seigniorage calculation audit trail

8. Troubleshooting Guide

8.1 'Analysis Failed' - Common Causes

Error: Fewer than 5 accepted images
Cause: User skipped too many images

Solution: Retake images, ensure each passes validation

Error: Granite detection model not configured
Cause: AI models not uploaded

Solution: Upload granite detection model to /models/hub/

Error: Measurement calibration required
Cause: No valid calibration found

Solution: Use correct calibration method for conditions

Error: Geometry reconstruction failed
Cause: Insufficient feature matches

Solution: Capture more images, ensure good overlap

8.2 'Validation Failed' - Common Issues

Issue: Duplicate image detected
Cause: Same file uploaded twice

Solution: Use different image or resize/rotate slightly

Issue: Not a Granite Block
Cause: AI detector doesn't see granite

Solution: Ensure granite clearly visible, fill frame

Issue: Blur detected
Cause: Camera movement during capture

Solution: Use both hands, steady camera

Issue: Weak feature overlap
Cause: Image too different from existing

Solution: Maintain 30-50% overlap with adjacent views

9. Configuration & Customization

9.1 Calibration Configuration

Backend settings (app/core/config.py):

```
# Detection thresholds granite_detection_confidence: float = 0.6
validation_quality_threshold: float = 0.5 # Capture requirements
minimum_evidence_images: int = 5 minimum_side_coverage: float = 0.8 # Measurement
tolerances maximum_dimension_variance_pct: float = 5.0
minimum_measurement_confidence: float = 0.7
```

9.2 Seigniorage Rates Configuration

Database table with classifications:

- GANGSAW_CLASS_1: Large blocks (>270cm length, >150cm breadth)
- GANGSAW_CLASS_2: Medium blocks
- GANGSAW_CLASS_3: Small blocks

10. Conclusion

10.1 Summary: Why This Matters for Departments

BEFORE GraniteMeasure AI:

✗ Manual measurements (tape measure, error-prone)

✗ Subjective disputes over dimensions

✗ No digital audit trail

✗ Revenue leakage through under-reporting

✗ Cutting decisions based on guesswork

AFTER GraniteMeasure AI:

✓ Digital twin creation (immutable evidence)

✓ Centimeter-level accuracy (1-2% typical)

✓ Complete audit trail (all actions logged)

✓ Automated seigniorage calculation

✓ Cutting optimization (maximize yield)

10.2 Key Metrics for Success

• Measurement Accuracy: ±2% (achievable with proper capture)

• Processing Time: <2 minutes per block (end-to-end)

• First-Time Pass Rate: >85% (with training)

• Seigniorage Recovery: 100% (no unmeasured blocks)

• Dispute Resolution: <24 hours (digital evidence)

10.3 Next Steps

• _WEEK 1_: Training, Pilot (10-20 blocks), Calibration setup

• _MONTH 1_: Optimization, Expansion to all quarries, Integration

• _QUARTER 1_: Advanced features, Native app, Reporting, Blockchain

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